

Beckett O'Brien

bobrien@college.harvard.edu ❖ (424) 355-6693 ❖ Cambridge, MA ❖ github.com/BeckettOBrien

EDUCATION

Harvard University

Sep. 2024 – May 2028

Major: Electrical Engineering, Minor: Computer Science

Cambridge, MA

- GPA: 3.96/4.0
- Relevant coursework: Computer Architecture, Digital Logic, Systems Programming, Operating Systems, Circuits, Signals, Control Systems, Discrete Math, Linear Algebra

RESEARCH & WORK EXPERIENCE

Harvard University - Architecture, Circuits, and Compilers Group

Jun. 2026 – Present

Undergraduate Research Assistant, advised by Prof. David Brooks and Prof. Gu-Yeon Wei

Cambridge, MA

- Designed and verified floating-point arithmetic units in both SystemC and Verilog across reduced-precision number formats (FP16, BF16, NVFP4, MXFP4, and more), quantifying the accuracy and PPA tradeoffs of each hardware implementation
- Built Python verification environments in cocotb to validate unit correctness against reference implementations and characterize numerical error across formats
- Synthesized and deployed modules on FPGAs using Vivado to validate correctness, timing, and resource utilization; used Yosys and Catapult HLS synthesis flows to estimate ASIC area, timing, and power
- Investigated agentic AI workflows for hardware design, using LLM agents to generate and iterate on floating-point unit RTL & HLS, and benchmarked the results against hand-written implementations

Harvard University, CS 1410: Computing Hardware

Feb. 2025 – Present

Course Assistant for Prof. Woodward Yang

Cambridge, MA

- Write SystemVerilog testbenches and Python tooling that interface with FPGAs and automate debugging, testing, and grading both in Vivado's simulator and on physical hardware
- Develop exam questions and lab materials covering digital logic, boolean algebra, CMOS, finite state machines, pipelining, memory hierarchy, CPU architecture, and FPGA design
- Guide students through digital logic design, hardware implementation, and waveform analysis in SystemVerilog and Vivado; conduct individualized design reviews to assess understanding of students' own implementations

Harvard Undergraduate Robotics Club

Sep. 2024 – Present

Software Team Lead, Electrical Team Co-Lead

Cambridge, MA

- Wrote firmware and low-level control software to drive a 6-DOF arm through precise, complex motion profiles using live sensor and motor feedback
- Architected the rover's distributed compute system over ROS2 and CANopen; designed power distribution, sensor, and communication PCBs

Survios

Jun. – Aug. 2024 & 2025

Software Engineering Intern

Los Angeles, CA

- Designed and implemented production-ready C++ subsystems (including hardware and network API integrations) that shipped in the final game

RELEVANT PROJECTS

- **MIPS CPU:** Designed and implemented a multi-cycle 32-bit MIPS CPU in SystemVerilog, synthesized and deployed on FPGA using Vivado; wrote comprehensive automated testbenches to validate correctness across the instruction set
- **NIC Driver/Network Stack:** Wrote a fully custom driver for the RTL8139 NIC and a network stack with ARP (client & server) and UDP socket abstractions for the Chickadee learning OS in C++ and Assembly
- **CSP Controller:** Built a cyclic synchronous positioning controller on a microcontroller commanding a CANopen motor network at a precise 1 kHz update rate; implemented a custom C++ Linux driver to stream interpolated position/velocity over UART with message integrity checks and error recovery

TECHNICAL SKILLS

- **Digital Design & Verification:** RTL/HLS, testbench development, waveform debugging, FPGA synthesis, timing analysis
- **Languages:** C/C++, Python, SystemVerilog/Verilog, SystemC, Assembly, Tcl, Bash
- **Circuits & Embedded:** PCB design, microcontroller firmware, hardware debugging
- **Protocols:** Ethernet, ARP, UDP, UART, SPI, I2C, CAN/CANopen
- **Tools:** Vivado, Yosys, Catapult HLS, cocotb, Linux, Git, Make, CMake